MicroGenius LE – Boost and HELIX Operation (Rev. P)

SENS Proprietary and Confidential Information

All charge voltages shall be as measured at the cable terminals and require voltage compensation to correct for cable losses

Initial Power-On
Boost Timeout = False
Battery LED = Off
Charger LED = Amber
(1 second minimum)
State = Boost
Start Boost Timer
Initialize Dynamic Boost Counter
Battery Health Counter = 15 minutes
Battery Recovery = Not Run
State Selection

Eco Float

Output < 0V? — Yes: Set Output = 0V, Charger LED = Flash Red/Yellow
Battery Connected? — No: Battery LED = Flash Red, Battery Connected = False, Battery Health Counter = 0
Battery Connected = False (previous state)? — Yes: Battery Connected = True
Charger LED = Green
24V Model? — Yes: Battery Health Timer > 0? — No: Output Voltage = 25.6; Yes: Output Voltage = 24.6
24V Model? — No: Battery Health Timer > 0? — Yes: Output Voltage = 12.3V; No: Output Voltage = 12.8V
Output Voltage = 24.6 → Output Voltage < 24.8V for > 100s? — No; Yes: Battery Recovery = Expired? — Yes: Battery LED = Red; No
Output Voltage = 12.3V → Output Voltage < 12.4V for > 100s? — Yes; No
Output Current > 95% of Current Rating? — Yes: State = Boost
Previous State = Eco Float? — Yes: Battery Recovery = Started, Battery LED = Flash Green
Start Eco Float Timer
Eco Float Timer > 10 Hours? — Yes: State = Refresh
AC Good? — No: Charger LED = Red, Battery LED = Off
AC Fail? — No

Refresh

Output < 0V? — Yes: Set Output = 0V, Charger LED = Flash Red/Yellow
Battery Connected? — No: Battery LED = Flash Red, Battery Connected = False, Battery Health Counter = 0
Charger LED = Flash 2x Short Green
24V Model? — Yes: Output Voltage = 28.6V
24V Model? — No: Output Voltage = 14.3V
Previous State = Refresh? — Yes; No: Start Refresh Timer
Refresh Timer > 2 Hours? — Yes: Battery Recovery = Started
Refresh Timer > 36 Hours? — Yes: Battery Recovery = Expired, Battery Health Counter = 15 minutes, Battery Health Timer Countdown Started (counts down to 0)
Output Current > 95% of Current Rating? — Yes: State = Boost; No: State = Eco Float

Boost

Output < 0V? — Yes: Set Output = 0V, Charger LED = Flash Red/Yellow
Battery Connected? — No: Battery LED = Flash Red, Battery Connected = False, Battery Health Counter = 0
Charger LED = Regular Flash Green
Boost Timeout = True? — Yes: Battery LED = Flash Amber
24V Model? — Yes: Output Voltage = 29V
24V Model? — No: Output Voltage = 14.5V
Output Current > 95% of Current Rating? — No: Output Current < 85% of Current Rating? — Yes: Decrement Dynamic Boost Counter by 2; No: Increment Dynamic Boost Counter by 1
Previous State = Boost? — Yes; No: Start Boost Timer
Initialize Dynamic Boost Counter
Battery Health Counter = 15 minutes
Dynamic Boost Counter = 0? — Yes: Battery Health Timer Countdown Started (counts down to 0), Battery LED = Green
Boost Timer > 36 Hours? — Yes: Boost Timeout = True, Battery LED = Flash Amber, State = Eco Float